LAB 5

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```
test> use college_UG_02_BTCSEAIML_2023_063
switched to db college_UG_02_BTCSEAIML_2023_063
college_UG_02_BTCSEAIML_2023_063> db.createCollection("student_UG_02_BTCSEAIML_2023_063")
{ ok: 1 }
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.insertMany([
... { emp_id: 101, name: "Amit", department: "IT", designation: "Developer", city: "Kolkata", salary: 60000, experience: 3 },
... { emp_id: 102, name: "Neha", department: "HR", designation: "HR Executive", city: "Delhi", salary: 45000, experience: 2 },
... { emp_id: 103, name: "Ravi", department: "IT", designation: "Tester", city: "Mumbai", salary: 50000, experience: 4 },
... { emp_id: 104, name: "Pooja", department: "Finance", designation: "Accountant", city: "Chennai", salary: 55000, experience: 5 },
... { emp_id: 105, name: "Karan", department: "IT", designation: "Developer", city: "Kolkata", salary: 65000, experience: 6 },
... { emp_id: 106, name: "Sneha", department: "HR", designation: "Manager", city: "Delhi", salary: 70000, experience: 7 },
... { emp_id: 107, name: "Arjun", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 48000, experience: 2 },
... { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
... { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
... { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b788d7346284aa08628ca0'),
    '1': ObjectId('69b788d7346284aa08628ca1'),
    '2': ObjectId('69b788d7346284aa08628ca2'),
    '3': ObjectId('69b788d7346284aa08628ca3'),
    '4': ObjectId('69b788d7346284aa08628ca4'),
    '5': ObjectId('69b788d7346284aa08628ca5'),
    '6': ObjectId('69b788d7346284aa08628ca6'),
    '7': ObjectId('69b788d7346284aa08628ca7'),
    '8': ObjectId('69b788d7346284aa08628ca8'),
    '9': ObjectId('69b788d7346284aa08628ca9')
  }
```

Q1. Display all employee records.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find()
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
```

```
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca6'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca9'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
```

Q2. Create an index on emp_id field.

Q3. Create an index on department field.

Q4. Create a compound index on department and salary.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.createIndex({emp_id:1})
emp_id_1
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.createIndex({department:1})
department_1
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.createIndex({department:1,salary:1})
department_1_salary_1
```

Q5. Find employees working in IT department.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({department:"IT"})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q6. Display employees whose salary is greater than 60,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({salary:{$gt:60000}})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q7. Sort employees by salary in descending order.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find().sort({salary:-1})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
```

Q8. Display top 3 highest paid employees.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find().sort({salary:-1}).limit(3)
[
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  }
]
```

Q9. Display employees having experience more than 5 years.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({experience:{$gt:5}})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q10. Analyze query performance using explain().

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({department:"IT"}).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'college_UG_02_BTCSEAIML_2023_063.student_UG_02_BTCSEAIML_2023_063',
    parsedQuery: { department: { '$eq': 'IT' } },
    indexFilterSet: false,
    queryHash: 'EB9C5E66',
    planCacheShapeHash: 'EB9C5E66',
    planCacheKey: '25CB5D62',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { department: 1 },
        indexName: 'department_1',
        isMultiKey: false,
        multiKeyPaths: { department: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { department: [ '["IT", "IT"]' ] }
      }
    },
    rejectedPlans: [
      {
        isCached: false,
        stage: 'FETCH',
        inputStage: {
          stage: 'IXSCAN',
          keyPattern: { department: 1, salary: 1 },
          indexName: 'department_1_salary_1',
          isMultiKey: false,
          multiKeyPaths: { department: [], salary: [] },
          isUnique: false,
          isSparse: false,
          isPartial: false,
          indexVersion: 2,
          direction: 'forward',
          indexBounds: {
            department: [ '["IT", "IT"]' ],
            salary: [ '[MinKey, MaxKey]' ]
          }
        }
      }
```